

CHAIYAPHUM, Thailand

WMO: 484030

Lat: 15.812N

Lon: 102.029E

Elev: 605

StdP: 14.38

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	60.4	63.1	46.8	48.3	69.6	50.2	55.1	70.4	11.9	77.6	11.0	78.0	1.4	50	0.310

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.7	101.0	77.5	99.0	76.9	97.1	76.6	81.4	93.6	80.3	91.1	79.7	89.9	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	151.6	85.8	77.9	148.2	84.6	77.3	145.2	83.5	45.7	93.8	44.5	91.4	43.8	90.2	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.7	7.4	55.4	103.5	3.5	2.0	52.8	104.9	50.7	106.1	48.8	107.2	46.2	108.7		
(5)			51.0	84.0	3.2	1.8	48.7	85.2	46.9	86.3	45.1	87.2	42.8	88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.3	77.1	80.9	84.9	87.2	85.7	84.9	83.5	82.8	82.4	79.8	76.7	(6)	
(7)		DBStd	4.69	4.29	4.79	4.44	3.68	3.24	2.41	2.48	2.11	2.05	2.34	3.52	4.67	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	2	1	0	0	0	0	0	0	0	0	0	0	1	(9)
(10)		CDD50	11801	839	866	1083	1117	1108	1046	1039	1016	973	991	895	828	(10)
(11)		CDD65	6327	374	446	618	667	643	596	574	551	523	526	445	364	(11)
(12)		CDH74	70388	3434	4926	7773	8722	7834	7151	6389	5818	5373	5448	4289	3231	(12)
(13)	CDH80	31601	1390	2450	4239	4858	3823	3236	2598	2159	1903	2054	1692	1199	(13)	
(14)	Wind	WSAvg	2.1	2.0	1.9	2.0	1.8	1.8	2.3	2.5	2.2	1.5	1.9	2.3	2.4	(14)
(15)	Precipitation	PrecAvg	47.5	0.2	0.7	2.0	3.5	6.5	5.6	5.9	5.5	11.2	5.5	0.8	0.2	(15)
(16)		PrecMax	65.7	0.9	3.8	4.1	6.9	10.1	9.4	12.4	10.9	24.4	14.4	4.3	1.4	(16)
(17)		PrecMin	31.9	0.0	0.0	0.0	1.6	0.3	1.2	2.4	1.9	4.1	0.6	0.0	0.0	(17)
(18)		PrecStd	10.1	0.3	1.0	1.2	1.6	3.0	2.4	3.1	2.3	5.8	3.4	1.1	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.9	99.4	102.6	104.3	101.7	98.0	97.3	94.3	93.4	93.2	95.0	93.6	(19)
(20)			MCWB	72.4	73.6	75.8	79.3	81.5	78.7	78.2	78.3	78.9	77.8	75.6	74.1	(20)
(21)		2%	DB	92.3	97.1	100.3	101.8	98.7	95.3	93.9	92.1	91.4	91.4	92.4	91.1	(21)
(22)			MCWB	72.6	73.4	75.4	78.3	79.4	78.4	77.9	78.1	78.6	77.0	75.6	73.1	(22)
(23)		5%	DB	90.0	95.0	98.2	99.4	96.0	93.5	91.7	90.4	89.9	90.0	90.1	88.9	(23)
(24)			MCWB	72.2	73.5	75.4	77.8	79.1	78.3	77.7	78.0	78.2	76.5	74.8	72.6	(24)
(25)		10%	DB	87.6	92.3	95.9	96.9	93.5	91.6	89.8	88.8	88.3	88.7	88.2	86.6	(25)
(26)			MCWB	71.3	73.3	75.4	77.5	78.9	78.2	77.6	77.8	77.8	76.2	73.9	71.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.1	78.0	81.1	82.8	83.9	81.2	80.6	80.7	80.8	80.3	78.8	76.8	(27)
(28)			MCDWB	89.0	93.1	95.8	100.9	97.2	92.1	90.9	88.7	89.0	88.8	88.4	87.8	(28)
(29)		2%	WB	75.4	76.4	78.9	80.3	81.8	80.1	79.5	79.7	79.8	79.2	77.7	75.4	(29)
(30)			MCDWB	87.1	91.0	93.8	95.9	93.9	90.3	89.0	87.7	87.7	87.1	87.1	86.5	(30)
(31)		5%	WB	74.0	75.5	77.6	79.2	80.3	79.4	78.8	79.0	79.1	78.4	76.8	74.0	(31)
(32)			MCDWB	85.5	89.6	92.2	93.9	91.3	89.2	87.7	87.1	86.6	85.9	86.0	85.5	(32)
(33)		10%	WB	72.7	74.5	76.6	78.5	79.4	78.9	78.2	78.3	78.6	77.8	75.7	72.8	(33)
(34)			MCDWB	84.1	88.0	90.7	92.3	89.7	88.2	86.6	86.0	85.7	84.8	84.3	84.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.5	21.4	20.3	18.7	15.2	13.8	13.0	12.5	12.1	13.1	15.9	18.4	(35)
(36)			MCDBR	23.3	24.1	23.5	21.6	18.4	16.1	15.5	14.8	14.3	14.7	17.9	20.2	(36)
(37)			MCWBR	7.7	6.9	6.5	5.6	4.9	3.9	3.9	4.1	4.2	4.2	5.3	6.4	(37)
(38)		5% WB	MCDBR	19.7	21.1	21.0	19.5	16.5	14.6	13.6	13.4	12.9	13.1	15.4	18.1	(38)
(39)			MCWBR	7.4	6.5	6.9	5.9	5.6	4.3	4.5	4.4	4.4	4.4	5.2	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.488	0.593	0.654	0.606	0.507	0.458	0.446	0.454	0.497	0.540	0.479	0.472	(40)	
(41)		taud	2.052	1.804	1.704	1.856	2.170	2.322	2.355	2.319	2.184	2.026	2.169	2.127	(41)	
(42)		Ebn at Noon	253	232	223	234	255	265	268	269	258	243	254	254	(42)	
(43)		Edh at Noon	51	69	78	67	48	41	40	42	48	54	45	47	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1544	1658	1700	1829	1761	1663	1564	1516	1480	1542	1556	1513	(44)	
(45)		RadStd	99	74	87	76	98	89	99	92	87	107	66	45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+284
(47)	Regional (26 neighbors)	+0.52	N/A	N/A	N/A	+0.38	+0.36	N/A	N/A	+232	+232

Nomenclature: See separate page